IDLE Shell 3.14.6

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```
Python 3.14.6 (tags/v3.14.6:c63aec6, Jun 10 2026, 10:26:10) [MSC v.1944 64 bit (AMD64)]
on win32
Enter "help" below or click "Help" above for more information.
>>>
======== RESTART: C:/Users/tthar/OneDrive/Documents/FODS/assessment 1.py =======
     Name  Age  Student
0    John   20     True
1    Emma   19     True
2    Liam   21    False
3  Olivia   18     True
>>>
```

assessment 1.py - C:/Users/tthar/OneDrive/Documents/FODS/assessment 1.py (3.14.6)

File Edit Format Run Options Window Help

```python
import pandas as pd

# Create DataFrame
df = pd.DataFrame()

# Add columns
df['Name'] = ['John', 'Emma', 'Liam', 'Olivia']
df['Age'] = [20, 19, 21, 18]
df['Student'] = [True, True, False, True]

# Display DataFrame
print(df)
```

assessment 10.py - C:/Users/tthar/OneDrive/Documents/FODS/assessment 10.py (3.14.6)

File Edit Format Run Options Window Help

```python
from sklearn.datasets import load_iris
from sklearn.cluster import KMeans

# Load Iris dataset
iris = load_iris()

# Create KMeans model
kmeans = KMeans(n_clusters=3)

# Train the model
kmeans.fit(iris.data)

# Get cluster labels
cluster_labels = kmeans.labels_

# Display cluster labels
print("Cluster Labels:", cluster_labels)
```

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```
1    Emma   19     True
2    Liam   21    False
3  Olivia   18     True
>>>
============ RESTART: C:/Users/tthar/OneDrive/Documents/FODS/assessment 2.py ===========
     Name  Age  Student
0    John   20     True
1    Emma   19     True
2    Liam   21    False
3  Olivia   18     True
4  Sophia   22    False
>>>
============ RESTART: C:/Users/tthar/OneDrive/Documents/FODS/assessment 3.py ===========
Mean: 30.0
Median: 30.0
Mode: ModeResult(mode=np.int64(10), count=np.int64(1))
>>>
============ RESTART: C:/Users/tthar/OneDrive/Documents/FODS/assessment 4.py ===========
Probability: 0.9331927987311419
>>>
============ RESTART: C:/Users/tthar/OneDrive/Documents/FODS/assessment 5.py ===========
T-Statistic: -0.2049800154226962
P-Value: 0.8456711339658733
>>>
============ RESTART: C:/Users/tthar/OneDrive/Documents/FODS/assessment6.py ============
Correlation: 1.0
P-Value: 0.0
>>>
============ RESTART: C:/Users/tthar/OneDrive/Documents/FODS/assessment 7.py ===========
Solution: [0.  2.5]
>>>
============ RESTART: C:/Users/tthar/OneDrive/Documents/FODS/assessment 8.py ===========
[-2.62955131e-08]
>>>
============ RESTART: C:/Users/tthar/OneDrive/Documents/FODS/assessment 9.py ===========
Accuracy: 1.0
              precision    recall  f1-score   support

           0       1.00      1.00      1.00        19
           1       1.00      1.00      1.00        13
           2       1.00      1.00      1.00        13

    accuracy                           1.00        45
   macro avg       1.00      1.00      1.00        45
weighted avg       1.00      1.00      1.00        45

>>>
=========== RESTART: C:/Users/tthar/OneDrive/Documents/FODS/assessment 10.py ===========
Cluster Labels: [1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
1
 1 1 1 1 1 1 1 1 1 1 1 1 1 0 2 0 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2
 2 2 2 0 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 0 2 0 0 0 0 2 0 0 0 0
 0 0 2 2 0 0 0 0 2 0 2 0 2 0 0 2 2 0 0 0 0 0 2 0 0 0 0 2 0 0 0 2 0 0 0 2 0
 0 2]
>>>
```

assessment 9.py - C:/Users/tthar/OneDrive/Documents/FODS/assessment 9.py (3.14.6)

File Edit Format Run Options Window Help

```python
import numpy as np
import matplotlib.pyplot as plt
from sklearn import datasets
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score, classification_report

# Load Iris dataset
iris = datasets.load_iris()

X = iris.data
y = iris.target

# Split the dataset
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=42
)

# Feature Scaling
scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)

# Create Logistic Regression model
log_reg = LogisticRegression()

# Train the model
log_reg.fit(X_train, y_train)

# Predict
y_pred = log_reg.predict(X_test)

# Calculate Accuracy
accuracy = accuracy_score(y_test, y_pred)

print("Accuracy:", accuracy)

# Display Classification Report
print(classification_report(y_test, y_pred))
```

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```
Python 3.14.6 (tags/v3.14.6:c63aec6, Jun 10 2026, 10:26:10) [MSC v.1944 64 bit (AMD64)]
on win32
Enter "help" below or click "Help" above for more information.
>>>
======== RESTART: C:/Users/tthar/OneDrive/Documents/FODS/assessment 1.py =======
     Name  Age  Student
0    John   20     True
1    Emma   19     True
2    Liam   21    False
3  Olivia   18     True
>>>
============ RESTART: C:/Users/tthar/OneDrive/Documents/FODS/assessment 2.py ===========
     Name  Age  Student
0    John   20     True
1    Emma   19     True
2    Liam   21    False
3  Olivia   18     True
4  Sophia   22    False
>>>
============ RESTART: C:/Users/tthar/OneDrive/Documents/FODS/assessment 3.py ===========
Mean: 30.0
Median: 30.0
Mode: ModeResult(mode=np.int64(10), count=np.int64(1))
>>>
============ RESTART: C:/Users/tthar/OneDrive/Documents/FODS/assessment 4.py ===========
Probability: 0.9331927987311419
>>>
============ RESTART: C:/Users/tthar/OneDrive/Documents/FODS/assessment 5.py ===========
T-Statistic: -0.2049800154226962
P-Value: 0.8456711339658733
>>>
============ RESTART: C:/Users/tthar/OneDrive/Documents/FODS/assessment6.py ============
Correlation: 1.0
P-Value: 0.0
>>>
============ RESTART: C:/Users/tthar/OneDrive/Documents/FODS/assessment 7.py ===========
Solution: [0.  2.5]
>>>
============ RESTART: C:/Users/tthar/OneDrive/Documents/FODS/assessment 8.py ===========
[-2.62955131e-08]
>>>
============ RESTART: C:/Users/tthar/OneDrive/Documents/FODS/assessment 9.py ===========
Accuracy: 1.0
              precision    recall  f1-score   support

           0       1.00      1.00      1.00        19
           1       1.00      1.00      1.00        13
           2       1.00      1.00      1.00        13

    accuracy                           1.00        45
   macro avg       1.00      1.00      1.00        45
weighted avg       1.00      1.00      1.00        45

>>>
```

assessment 8.py - C:/Users/tthar/OneDrive/Documents/FODS/assessment 8.py (3.14.6)

File Edit Format Run Options Window Help

```python
from scipy.optimize import minimize

# Define the objective function
def objective(x):
    return x**2 + 4

# Find the minimum value
result = minimize(objective, x0=5)

# Display the optimal value
print(result.x)
```

Ln: 12 Col: 0

IDLE Shell 3.14.6

File Edit Shell Debug Options Window Help

```
Python 3.14.6 (tags/v3.14.6:c63aec6, Jun 10 2026, 10:26:10) [MSC v.1944 64 bit (AMD64)]
on win32
Enter "help" below or click "Help" above for more information.
>>>
======== RESTART: C:/Users/tthar/OneDrive/Documents/FODS/assessment 1.py =======
     Name  Age  Student
0    John   20     True
1    Emma   19     True
2    Liam   21    False
3  Olivia   18     True
>>>
============ RESTART: C:/Users/tthar/OneDrive/Documents/FODS/assessment 2.py ===========
     Name  Age  Student
0    John   20     True
1    Emma   19     True
2    Liam   21    False
3  Olivia   18     True
4  Sophia   22    False
>>>
============ RESTART: C:/Users/tthar/OneDrive/Documents/FODS/assessment 3.py ===========
Mean: 30.0
Median: 30.0
Mode: ModeResult(mode=np.int64(10), count=np.int64(1))
>>>
============ RESTART: C:/Users/tthar/OneDrive/Documents/FODS/assessment 4.py ===========
Probability: 0.9331927987311419
>>>
============ RESTART: C:/Users/tthar/OneDrive/Documents/FODS/assessment 5.py ===========
T-Statistic: -0.2049800154226962
P-Value: 0.8456711339658733
>>>
============ RESTART: C:/Users/tthar/OneDrive/Documents/FODS/assessment6.py ============
Correlation: 1.0
P-Value: 0.0
>>>
============ RESTART: C:/Users/tthar/OneDrive/Documents/FODS/assessment 7.py ===========
Solution: [0.  2.5]
>>>
============ RESTART: C:/Users/tthar/OneDrive/Documents/FODS/assessment 8.py ===========
[-2.62955131e-08]
>>>
```

Ln: 40 Col: 0

assessment 7.py - C:/Users/tthar/OneDrive/Documents/FODS/assessment 7.py (3.14.6)

File Edit Format Run Options Window Help

```python
from scipy import linalg

# Coefficient matrix
A = [[3, 2],
     [1, 2]]

# Constant values
B = [5, 5]

# Solve the linear equations
solution = linalg.solve(A, B)

# Display the solution
print("Solution:", solution)
```

IDLE Shell 3.14.6

File Edit Shell Debug Options Window Help

```
Python 3.14.6 (tags/v3.14.6:c63aec6, Jun 10 2026, 10:26:10) [MSC v.1944 64 bit (AMD64)]
on win32
Enter "help" below or click "Help" above for more information.
>>>
======== RESTART: C:/Users/tthar/OneDrive/Documents/FODS/assessment 1.py =======
     Name  Age  Student
0    John   20     True
1    Emma   19     True
2    Liam   21    False
3  Olivia   18     True
>>>
============ RESTART: C:/Users/tthar/OneDrive/Documents/FODS/assessment 2.py ===========
     Name  Age  Student
0    John   20     True
1    Emma   19     True
2    Liam   21    False
3  Olivia   18     True
4  Sophia   22    False
>>>
============ RESTART: C:/Users/tthar/OneDrive/Documents/FODS/assessment 3.py ===========
Mean: 30.0
Median: 30.0
Mode: ModeResult(mode=np.int64(10), count=np.int64(1))
>>>
============ RESTART: C:/Users/tthar/OneDrive/Documents/FODS/assessment 4.py ===========
Probability: 0.9331927987311419
>>>
============ RESTART: C:/Users/tthar/OneDrive/Documents/FODS/assessment 5.py ===========
T-Statistic: -0.2049800154226962
P-Value: 0.8456711339658733
>>>
============ RESTART: C:/Users/tthar/OneDrive/Documents/FODS/assessment6.py ============
Correlation: 1.0
P-Value: 0.0
>>>
============ RESTART: C:/Users/tthar/OneDrive/Documents/FODS/assessment 7.py ===========
Solution: [0.  2.5]
>>>
```

assessment6.py - C:/Users/tthar/OneDrive/Documents/FODS/assessment6.py (3.14.6)

File Edit Format Run Options Window Help

```python
from scipy.stats import pearsonr

# Sample data
x = [1, 2, 3, 4, 5]
y = [2, 4, 6, 8, 10]

# Calculate Pearson Correlation
corr, p_value = pearsonr(x, y)

# Display result
print("Correlation:", corr)
print("P-Value:", p_value)
```

Ln: 13 Col: 0

IDLE Shell 3.14.6

File Edit Shell Debug Options Window Help

```
Python 3.14.6 (tags/v3.14.6:c63aec6, Jun 10 2026, 10:26:10) [MSC v.1944 64 bit (AMD64)]
on win32
Enter "help" below or click "Help" above for more information.
>>>
======== RESTART: C:/Users/tthar/OneDrive/Documents/FODS/assessment 1.py =======
     Name  Age  Student
0    John   20     True
1    Emma   19     True
2    Liam   21    False
3  Olivia   18     True
>>>
============ RESTART: C:/Users/tthar/OneDrive/Documents/FODS/assessment 2.py ===========
     Name  Age  Student
0    John   20     True
1    Emma   19     True
2    Liam   21    False
3  Olivia   18     True
4  Sophia   22    False
>>>
============ RESTART: C:/Users/tthar/OneDrive/Documents/FODS/assessment 3.py ===========
Mean: 30.0
Median: 30.0
Mode: ModeResult(mode=np.int64(10), count=np.int64(1))
>>>
============ RESTART: C:/Users/tthar/OneDrive/Documents/FODS/assessment 4.py ===========
Probability: 0.9331927987311419
>>>
============ RESTART: C:/Users/tthar/OneDrive/Documents/FODS/assessment 5.py ===========
T-Statistic: -0.2049800154226962
P-Value: 0.8456711339658733
>>>
============ RESTART: C:/Users/tthar/OneDrive/Documents/FODS/assessment6.py ============
Correlation: 1.0
P-Value: 0.0
>>>
```

Ln: 34 Col: 0

assessment 5.py - C:/Users/tthar/OneDrive/Documents/FODS/assessment 5.py (3.14.6)

File Edit Format Run Options Window Help

```python
from scipy import stats

# Sample data
data = [22, 25, 19, 24, 28, 30]

# One-sample t-test
t_stat, p_value = stats.ttest_1samp(data, 25)

# Display results
print("T-Statistic:", t_stat)
print("P-Value:", p_value)
```

Ln: 12 Col: 0

IDLE Shell 3.14.6

File Edit Shell Debug Options Window Help

```
Python 3.14.6 (tags/v3.14.6:c63aec6, Jun 10 2026, 10:26:10) [MSC v.1944 64 bit (AMD64)]
on win32
Enter "help" below or click "Help" above for more information.
>>>
======== RESTART: C:/Users/tthar/OneDrive/Documents/FODS/assessment 1.py =======
     Name  Age  Student
0    John   20     True
1    Emma   19     True
2    Liam   21    False
3  Olivia   18     True
>>>
============ RESTART: C:/Users/tthar/OneDrive/Documents/FODS/assessment 2.py ===========
     Name  Age  Student
0    John   20     True
1    Emma   19     True
2    Liam   21    False
3  Olivia   18     True
4  Sophia   22    False
>>>
============ RESTART: C:/Users/tthar/OneDrive/Documents/FODS/assessment 3.py ===========
Mean: 30.0
Median: 30.0
Mode: ModeResult(mode=np.int64(10), count=np.int64(1))
>>>
============ RESTART: C:/Users/tthar/OneDrive/Documents/FODS/assessment 4.py ===========
Probability: 0.9331927987311419
>>>
============ RESTART: C:/Users/tthar/OneDrive/Documents/FODS/assessment 5.py ===========
T-Statistic: -0.2049800154226962
P-Value: 0.8456711339658733
>>>
```

Ln: 30 Col: 0

assessment 4.py - C:/Users/tthar/OneDrive/Documents/FODS/assessment 4.py (3.14.6)

File Edit Format Run Options Window Help

```python
from scipy.stats import norm

# Calculate cumulative probability
probability = norm.cdf(85, loc=70, scale=10)

# Display result
print("Probability:", probability)
```

Ln: 8 Col: 0

IDLE Shell 3.14.6

File Edit Shell Debug Options Window Help

```
Python 3.14.6 (tags/v3.14.6:c63aec6, Jun 10 2026, 10:26:10) [MSC v.1944 64 bit (AMD64)]
on win32
Enter "help" below or click "Help" above for more information.
>>> 
======== RESTART: C:/Users/tthar/OneDrive/Documents/FODS/assessment 1.py =======
     Name  Age  Student
0    John   20     True
1    Emma   19     True
2    Liam   21    False
3  Olivia   18     True
>>> 
============ RESTART: C:/Users/tthar/OneDrive/Documents/FODS/assessment 2.py ===========
     Name  Age  Student
0    John   20     True
1    Emma   19     True
2    Liam   21    False
3  Olivia   18     True
4  Sophia   22    False
>>> 
============ RESTART: C:/Users/tthar/OneDrive/Documents/FODS/assessment 3.py ===========
Mean: 30.0
Median: 30.0
Mode: ModeResult(mode=np.int64(10), count=np.int64(1))
>>> 
============ RESTART: C:/Users/tthar/OneDrive/Documents/FODS/assessment 4.py ===========
Probability: 0.9331927987311419
>>> 
```

Ln: 26 Col: 0

IDLE Shell 3.14.6

File Edit Shell Debug Options Window Help

```
Python 3.14.6 (tags/v3.14.6:c63aec6, Jun 10 2026, 10:26:10) [MSC v.1944 64 bit (AMD64)]
on win32
Enter "help" below or click "Help" above for more information.
>>>
======== RESTART: C:/Users/tthar/OneDrive/Documents/FODS/assessment 1.py =======
     Name  Age  Student
0    John   20     True
1    Emma   19     True
2    Liam   21    False
3  Olivia   18     True
>>>
============ RESTART: C:/Users/tthar/OneDrive/Documents/FODS/assessment 2.py ===========
     Name  Age  Student
0    John   20     True
1    Emma   19     True
2    Liam   21    False
3  Olivia   18     True
4  Sophia   22    False
>>>
============ RESTART: C:/Users/tthar/OneDrive/Documents/FODS/assessment 3.py ===========
Mean: 30.0
Median: 30.0
Mode: ModeResult(mode=np.int64(10), count=np.int64(1))
>>>
```

Ln: 23 Col: 0

assessment 3.py - C:/Users/tthar/OneDrive/Documents/FODS/assessment 3.py (3.14.6)

File Edit Format Run Options Window Help

```python
from scipy import stats
import numpy as np

# Sample data
data = [10, 20, 30, 40, 50]

# Mean
print("Mean:", np.mean(data))

# Median
print("Median:", np.median(data))

# Mode
print("Mode:", stats.mode(data))
```

Ln: 15 Col:

IDLE Shell 3.14.6

File Edit Shell Debug Options Window Help

```
Python 3.14.6 (tags/v3.14.6:c63aec6, Jun 10 2026, 10:26:10) [MSC v.1944 64 bit (AMD64)]
on win32
Enter "help" below or click "Help" above for more information.
>>>
======== RESTART: C:/Users/tthar/OneDrive/Documents/FODS/assessment 1.py =======
     Name  Age  Student
0    John   20     True
1    Emma   19     True
2    Liam   21    False
3  Olivia   18     True
>>>
============ RESTART: C:/Users/tthar/OneDrive/Documents/FODS/assessment 2.py ===========
     Name  Age  Student
0    John   20     True
1    Emma   19     True
2    Liam   21    False
3  Olivia   18     True
4  Sophia   22    False
>>>
```

Ln: 18 Col: 0

assessment 2.py - C:/Users/tthar/OneDrive/Documents/FODS/assessment 2.py (3.14.6)

File Edit Format Run Options Window Help

```python
import pandas as pd

# Create DataFrame
df = pd.DataFrame({
    'Name': ['John', 'Emma', 'Liam', 'Olivia'],
    'Age': [20, 19, 21, 18],
    'Student': [True, True, False, True]
})

# New row to add
new_row = pd.DataFrame(
    [['Sophia', 22, False]],
    columns=['Name', 'Age', 'Student']
)

# Add the new row
df = pd.concat([df, new_row], ignore_index=True)

# Display DataFrame
print(df)
```

Ln: 21 Col: 0